

WASHINGTON COUNTY AP, PA, USA

WMO: 725117

Lat: 40.133N

Lon: 80.283W

Elev: 361

StdP: 97.06

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04827

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-13.4	-22.1	0.5	-15.0	-18.9	0.7	-11.4	10.8	2.0	9.3	-0.8	2.0	240	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	31.3	22.1	29.8	21.6	28.5	20.9	23.8	28.7	22.9	27.6	22.1	26.9	3.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.8	27.0	21.3	16.7	25.6	20.9	16.2	25.1	72.8	28.8	69.4	27.6	66.2	26.9	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.5	6.5	-20.9	32.8	3.7	2.0	-23.6	34.2	-25.8	35.4	-27.9	36.5	-30.6	37.9		
			-21.4	25.3	3.9	1.0	-24.2	26.0	-26.5	26.6	-28.6	27.2	-31.4	28.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-2.1	-0.4	3.7	10.1	15.2	19.6	21.6	20.8	17.2	11.4	5.2	0.8
	DBStd	9.72	7.24	6.53	5.99	5.22	4.55	3.44	2.81	2.75	3.86	4.98	5.25	5.83
	HDD10.0	1451	381	297	211	62	8	0	0	0	1	41	160	290
	HDD18.3	3249	634	526	453	250	119	28	6	9	66	222	393	543
	CDD10.0	1571	5	4	17	66	169	288	360	336	218	86	17	5
	CDD18.3	326	0	0	1	3	22	66	108	86	32	8	0	0
	CDH23.3	2996	0	1	8	63	249	612	923	731	321	88	2	0
	CDH26.7	792	0	0	1	8	43	167	273	197	78	24	0	0
Wind	WSAvg	2.4	3.2	3.1	3.0	3.0	2.4	2.0	1.8	1.5	1.7	2.1	2.7	2.7
Precipitation	PrecAvg	1022	83	64	85	85	106	105	103	90	86	66	72	77
	PrecMax	1359	172	127	150	130	171	180	186	169	227	135	131	176
	PrecMin	822	44	29	37	37	43	37	52	47	36	21	19	42
	PrecStd	138	35	24	32	22	34	37	37	34	44	28	28	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.9	19.0	23.9	27.8	29.1	32.2	32.8	32.6	31.2	29.8	22.3	17.7
		MCWB	13.0	12.7	14.8	16.8	20.1	21.5	23.7	22.2	21.4	20.8	15.0	13.2
	2%	DB	14.9	16.1	21.0	25.2	27.7	30.1	31.2	30.9	28.8	25.9	19.1	15.9
		MCWB	11.6	11.1	12.8	15.3	18.8	21.5	22.5	22.3	20.9	18.6	13.1	12.4
	5%	DB	12.2	12.7	17.4	22.7	26.3	28.7	29.1	28.8	27.3	22.7	17.3	12.8
		MCWB	9.4	8.6	10.9	14.2	18.4	21.1	21.7	21.3	19.8	16.3	11.6	9.6
	10%	DB	8.0	9.0	13.8	20.1	24.0	27.3	27.8	27.5	25.0	20.3	14.1	10.9
		MCWB	5.5	6.0	9.0	12.9	17.5	20.2	20.9	20.4	18.5	15.6	10.1	8.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	14.6	15.7	18.6	22.3	24.0	25.5	24.3	23.7	22.0	16.4	14.9
		MCDB	17.2	17.8	22.4	25.7	27.4	29.3	31.3	29.9	28.1	28.0	19.8	17.1
	2%	WB	12.0	12.2	13.8	16.7	20.8	22.9	24.1	23.3	22.0	19.7	14.1	12.8
		MCDB	14.4	14.8	19.1	22.7	26.0	28.1	29.0	28.2	26.7	23.4	17.7	14.8
	5%	WB	9.4	9.1	11.9	15.2	19.7	22.1	22.9	22.4	20.8	17.8	12.4	10.5
		MCDB	11.7	12.8	15.7	20.5	24.5	27.3	27.3	27.0	25.0	21.7	16.2	12.4
	10%	WB	5.7	6.2	9.7	13.9	18.5	21.1	22.0	21.6	19.7	15.8	10.5	7.7
		MCDB	8.3	9.5	14.2	19.2	22.7	25.8	26.6	26.0	23.6	19.6	14.0	9.8
Mean Daily Temperature Range	5% DB	MDBR	9.1	10.1	11.4	13.2	12.7	12.0	11.7	12.0	12.8	12.1	10.6	8.9
		MCDBR	11.3	14.6	16.3	16.9	15.1	14.1	13.3	13.8	14.9	15.6	14.5	12.3
		MCWBR	8.9	10.8	10.2	9.3	8.3	7.2	6.5	6.8	7.8	9.1	9.9	9.8
	5% WB	MCDBR	10.5	13.2	13.8	13.9	12.3	11.8	11.3	11.2	12.1	13.0	12.5	11.0
		MCWBR	8.8	10.7	9.7	8.6	7.0	6.4	6.0	6.1	6.7	7.9	9.6	9.8
Clear-Sky Solar Irradiance	taub	0.302	0.323	0.348	0.380	0.425	0.456	0.467	0.469	0.395	0.363	0.320	0.300	
	taud	2.464	2.387	2.392	2.332	2.212	2.163	2.150	2.124	2.315	2.405	2.504	2.511	
	Ebn at Noon	866	893	905	894	856	827	813	799	843	834	838	842	
	Edh at Noon	78	97	108	123	141	149	150	149	116	95	75	70	
All-Sky Solar Radiation	RadAvg	1.55	2.40	3.38	4.56	5.29	5.76	5.72	5.23	4.29	2.90	1.98	1.33	
	RadStd	0.16	0.35	0.30	0.43	0.44	0.46	0.38	0.30	0.44	0.30	0.26	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130	+62

Nomenclature: See separate page